

JavaScript Data Types

What is a Data Type?

A **Data Type** tells JavaScript what kind of value a variable stores.

```
let name = "Rokon"; // String
let age = 23; // Number
```

Types of Data

JavaScript has **2 main types** of data:

1. Primitive Data Types
2. Non-Primitive (Reference) Data Types

1. Primitive Data Types

Primitive data types store a **single value** and are **immutable**.

There are **7 primitive data types**:

Data Type	Description	Example
String	Text	"Hello"
Number	Integer or Decimal	10, 3.14
Boolean	true or false	true
Undefined	Variable declared but not assigned	undefined
Null	Intentionally empty value	null
BigInt	Very large integer	123456789n
Symbol	Unique identifier	Symbol("id")

Examples

```
let name = "Rokon";
let age = 23;
let isStudent = true;
let city;
let car = null;
let bigNumber = 12345678901234567890n;
let id = Symbol("userId");
```

2. Non-Primitive (Reference) Data Types

Non-Primitive data types can store **multiple values** and are stored **by reference**.

Object

```
const student = {
  name: "Rokon",
  age: 23,
};

console.log(student.name);
```

Array

```
const fruits = ["Apple", "Mango", "Orange"];

console.log(fruits[0]);
```

Function

```
function greet() {  
  console.log("Hello!");  
}  
  
greet();
```

Primitive vs Non-Primitive

Feature	Primitive	Non-Primitive
Stores	Single Value	Multiple Values
Mutable	No	Yes
Copied By	Value	Reference
Examples	String, Number, Boolean	Object, Array, Function

Primitive Example

```
let a = 10;  
let b = a;  
  
b = 20;  
  
console.log(a); // 10  
console.log(b); // 20
```

Explanation: `b` gets a copy of `a`, so changing `b` does not affect `a`.

Non-Primitive Example

```
let person1 = {  
  name: "Rokon",  
};  
  
let person2 = person1;  
  
person2.name = "Rahim";  
  
console.log(person1.name); // Rahim  
console.log(person2.name); // Rahim
```

Explanation: Both variables point to the same object, so changing one also changes the other.

Easy Way to Remember

- **Primitive** = Each variable has its **own copy**.
- **Non-Primitive** = Multiple variables can point to the **same object**.

Interview Question

Q: What is the difference between Primitive and Non-Primitive Data Types?

Answer:

Primitive data types store a single value and are copied **by value**, while Non-Primitive data types store multiple values and are copied **by reference**.